Seminar 1
AI Research Study & Career Path: How to get started

AI VIET NAM
@aivietnam.edu.vn

AIMA
AI VIETNAM RESEARCH LAB

Nguyen Thanh Huy

Outline

- ❖ AI Researcher vs AI Engineer
- ❖ Academia Picture & Study Abroad
- ❖ How to find your own research direction

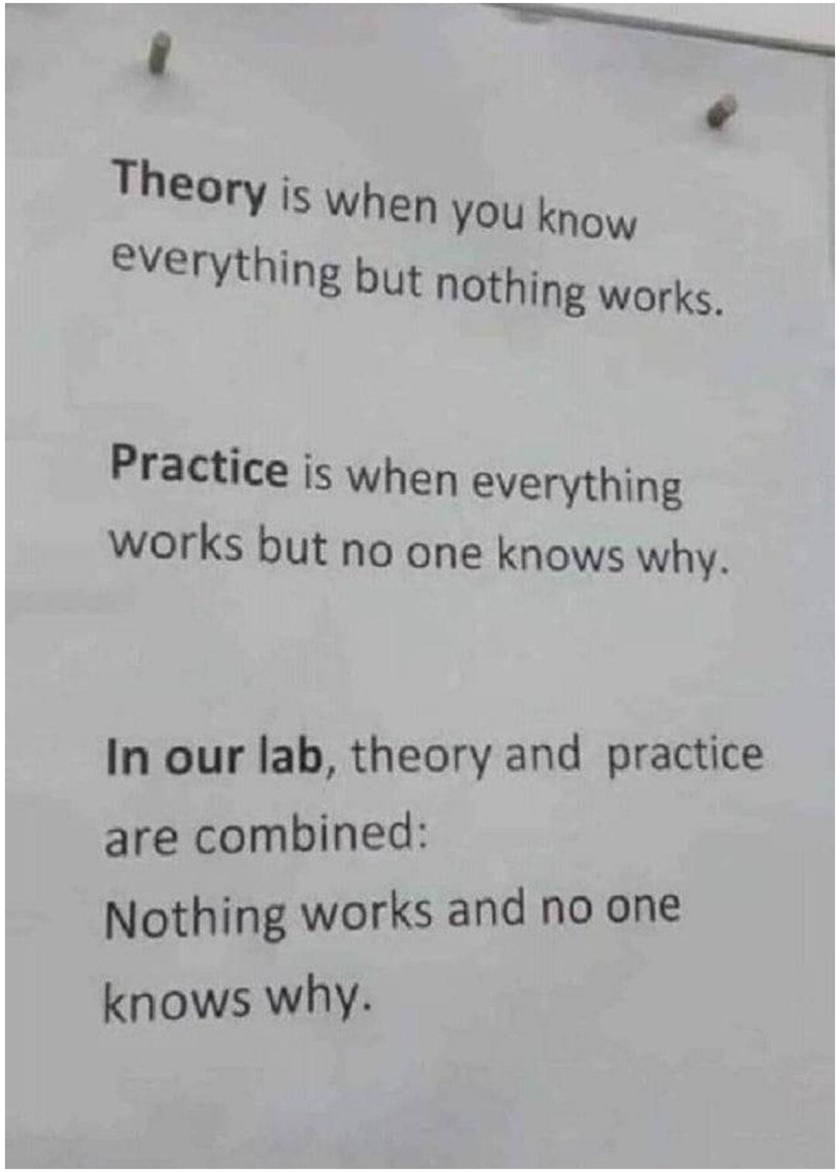
AI Researcher vs AI Engineer

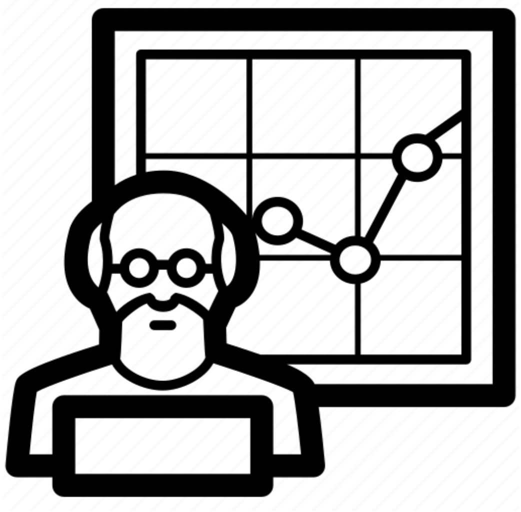
AI VIET NAM
@aivietnam.edu.vn

AIMA
AI VIETNAM RESEARCH LAB

Nguyen Thanh Huy

AI Researcher vs AI Engineer





VS

Comparision

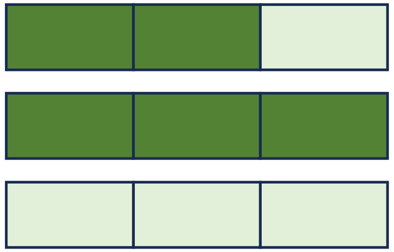
AI Engineer

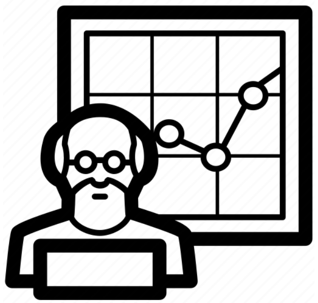
- Requirement: Mostly Bachelor or Master
- Career Development: Hybrid (Software, Data Scientist, Product Manager,…)
- Average Rating:

Time:

Income:

Freedom:





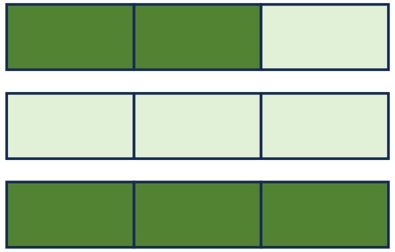
AI Researcher

- Requirement: Mostly PhD or Master
- Career Development: More Research-based
- Average Rating:

Time:

Income:

Freedom:



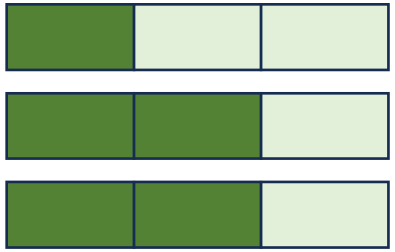
Professor

✓ Permanent contract

Time:

Income:

Freedom:



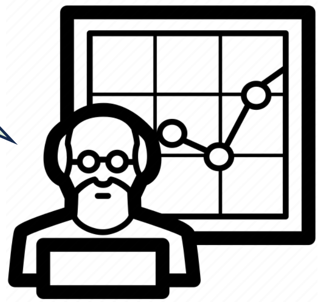
Research Scientist

AI Researcher vs AI Engineer

- Optimization
- Data-Centric
- Product & Cost

- Benchmarking
- Model-Centric
- Truth & Knowledge

AI Engineer



AI Researcher

However, mindsets & skillsets are different totally

Everyone might have already known what AI Engineers do
But today, let’s just talk about AI Researcher

Academia Picture & Study Abroad

Nguyen Thanh Huy

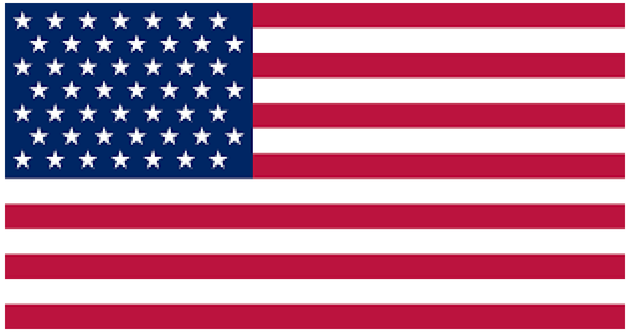
Duration for Studying Abroad

Europe

Master: 1-2 years
PhD: 3 years

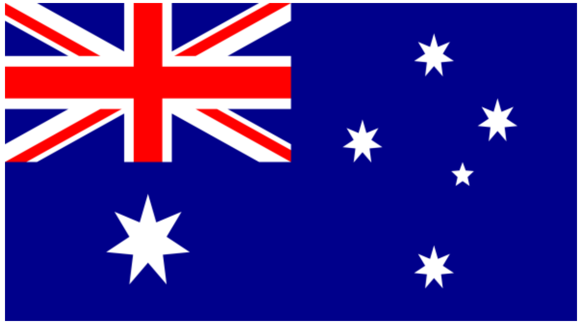
Asia

Master: 2 years
PhD: 3 years



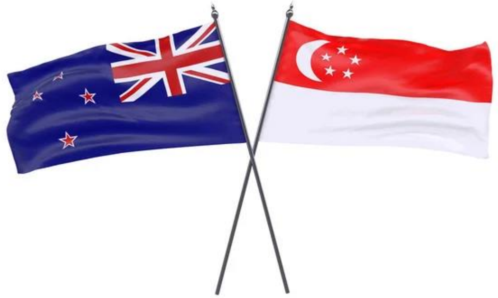
US

Master: 2 years (X)
PhD: 5 years (direct)



Australia

Master: 1-2 years
PhD: 3.5 years (direct)



New Zealand & Singapore

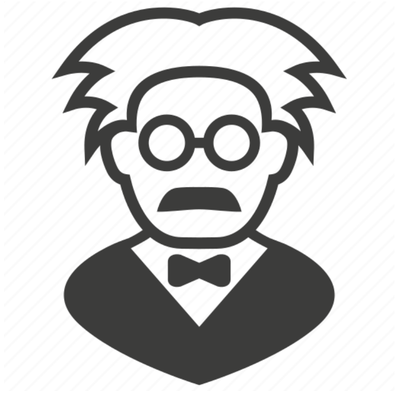
Master: 2 years
PhD: 4 years (direct)

Canada

Master: 2 years
PhD: 3 years

How does Academia work?

University

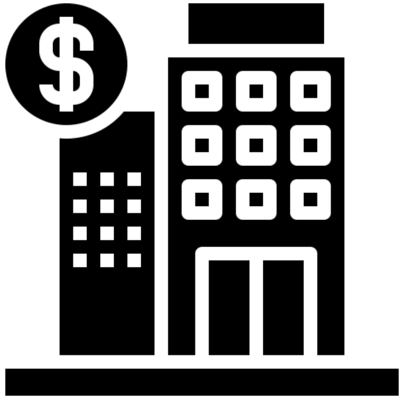


Professor/PI

Funding

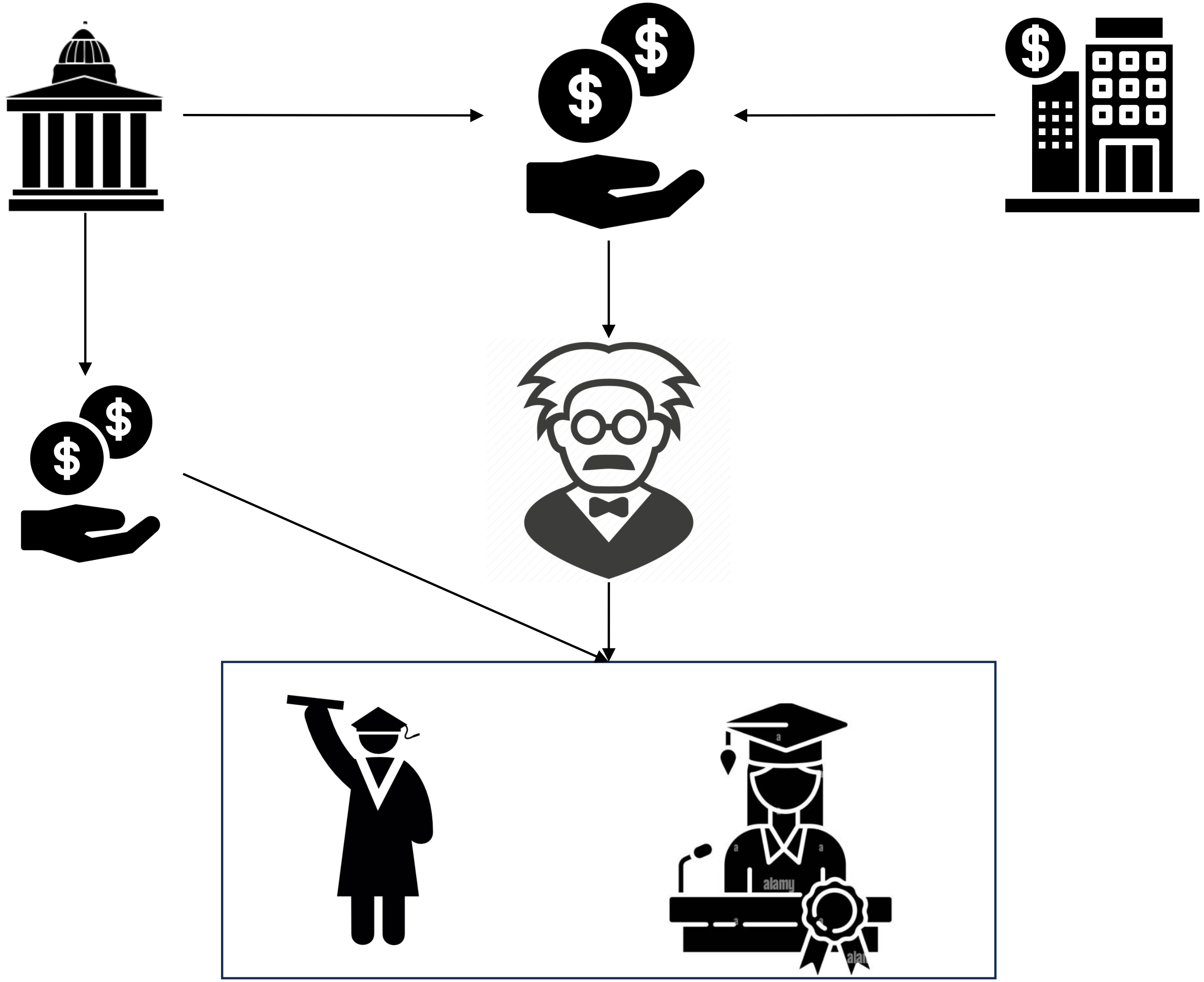
Master Degree

PhD Degree



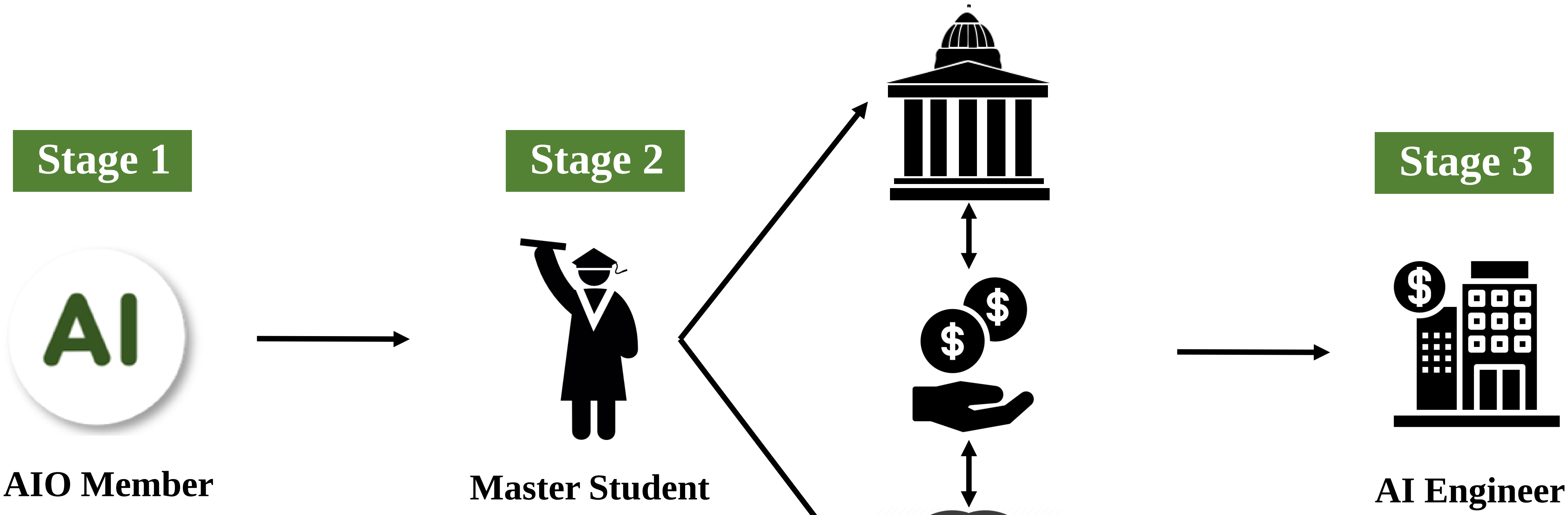
Company

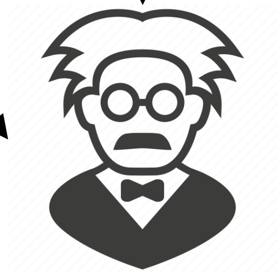
How does Academia work?



Scenario 1.1: Master Degree to Industry

- Asia (2 years)



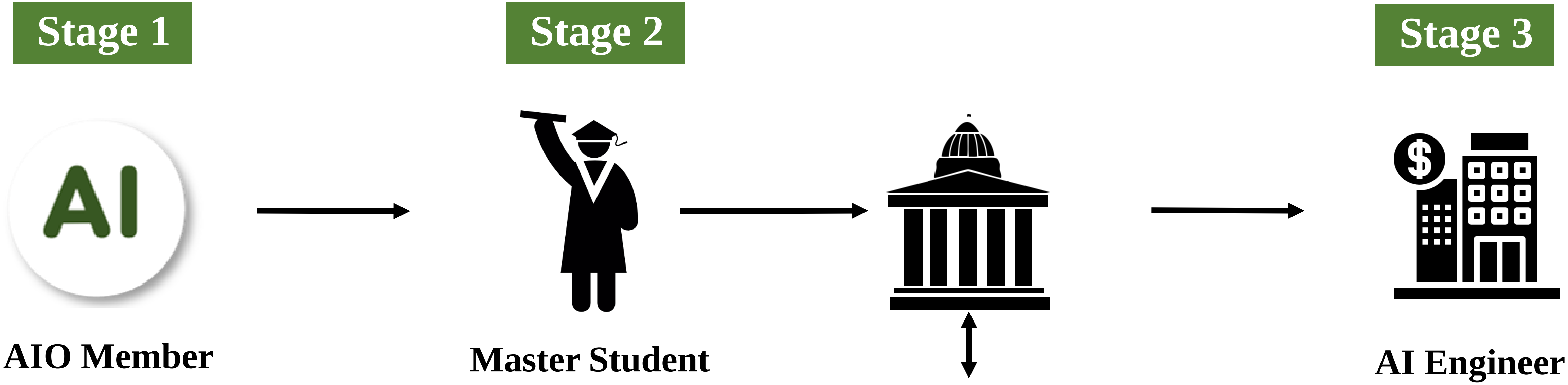


Requirement:

- Intermediate English (IELTS 5.5-6.0)
- Average Research Experience or 1 publication (any rank)
- Income is important. Life is flexible
- A Lot of scholarships

Scenario 1.2: Master Degree to Industry

- **Europe (1 or 2 years)**

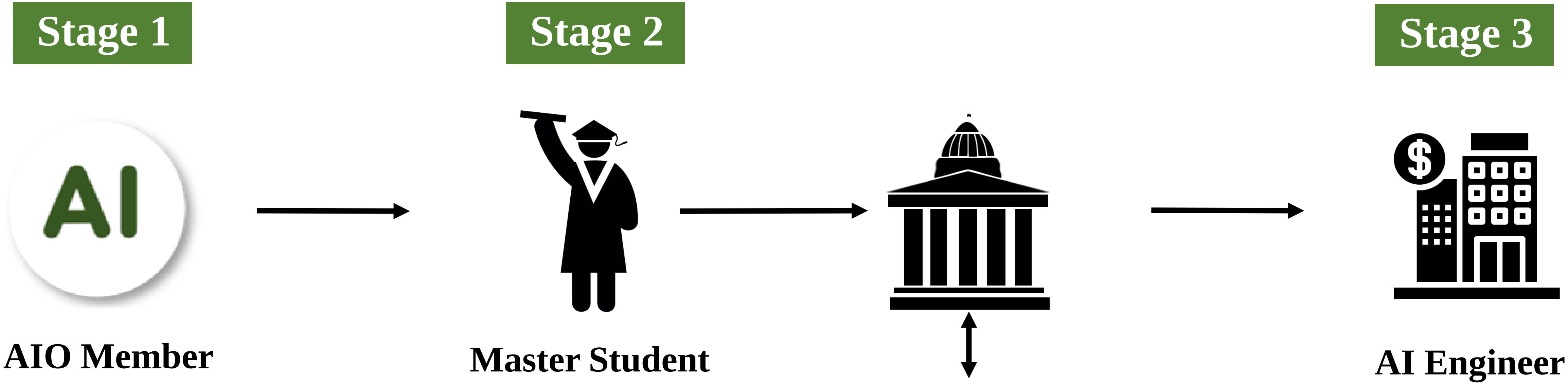


Requirement:

- Advanced English (IELTS 6.5-7.0+)
- No need research experience or paper
- Income is flexible. Life is important
- Few Scholarships available

Scenario 1.3: Master Degree to Industry

- **North America, Australia, UK (1 or 2 years) (AAU)**

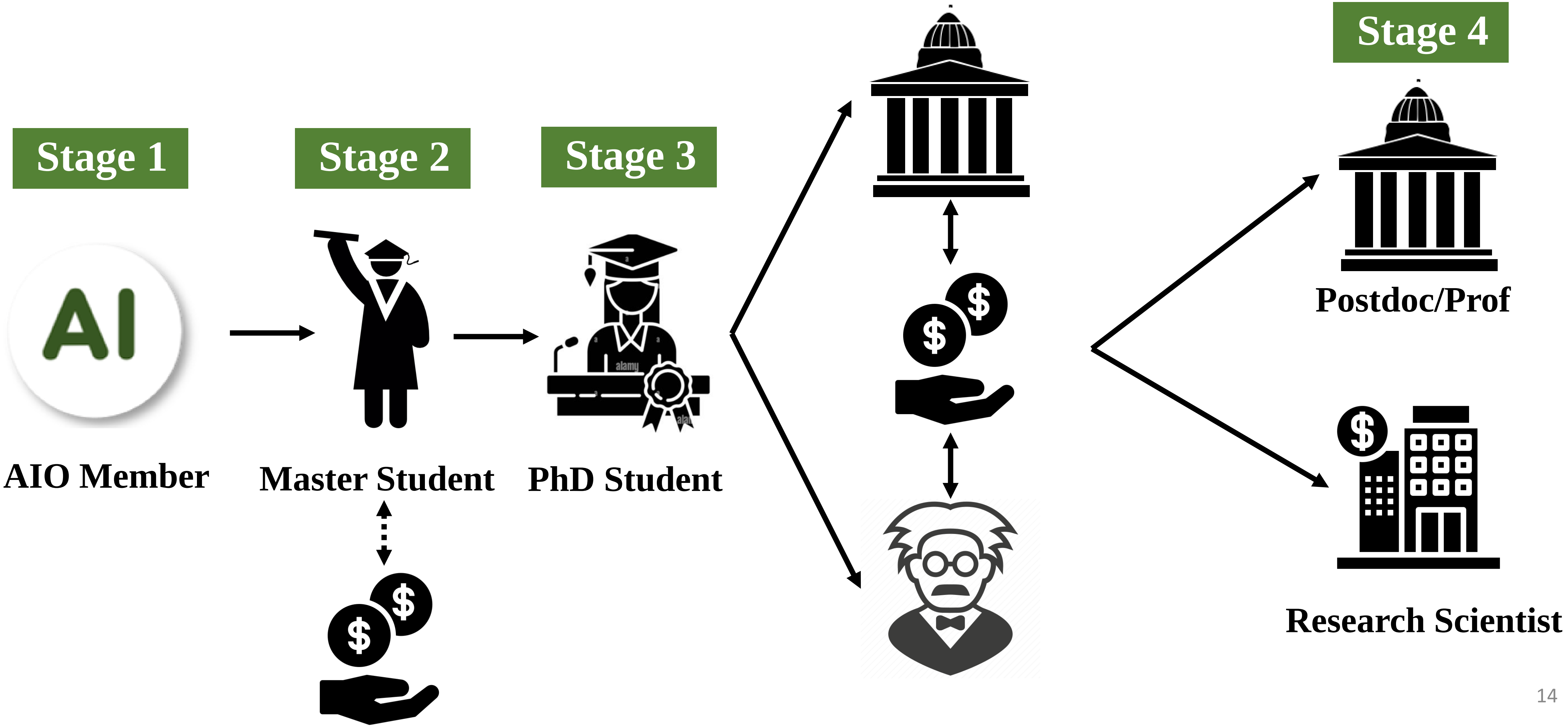


Requirement:

- Advanced English (IELTS 6.5-7.0+)
- Average Research Experience or 1 publication (any rank)
- Income is flexible. Life is important
- High-risk investment (No scholarship)

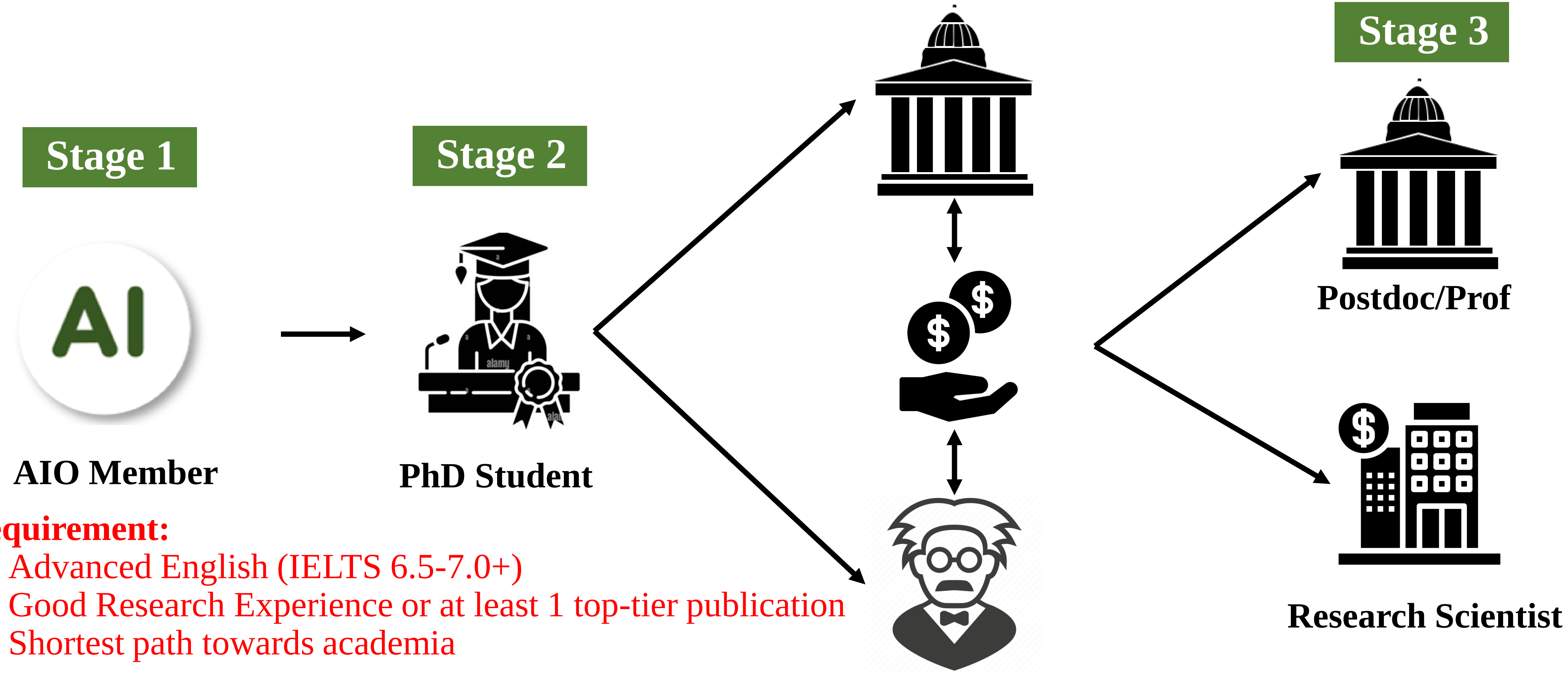
Scenario 2: Master to PhD

- Asia to Asia/Europe (5 years) or Asia to AAU(5-7 Years)



Scenario 3: Direct PhD

- **US, Australia, Singapore, Some EU Institutions (3-5 years)**



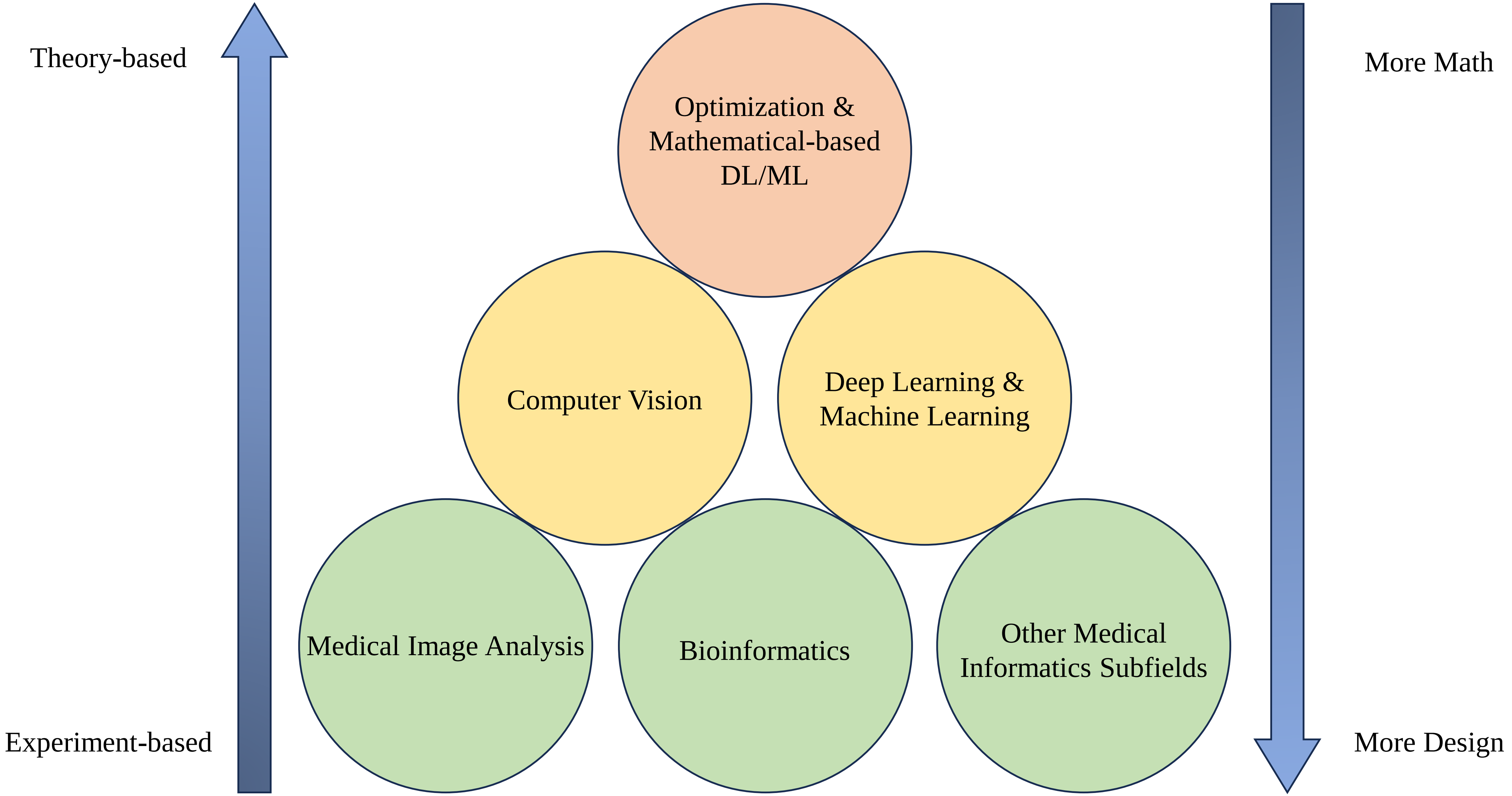
Requirement:

- Advanced English (IELTS 6.5-7.0+)
- Good Research Experience or at least 1 top-tier publication
- Shortest path towards academia

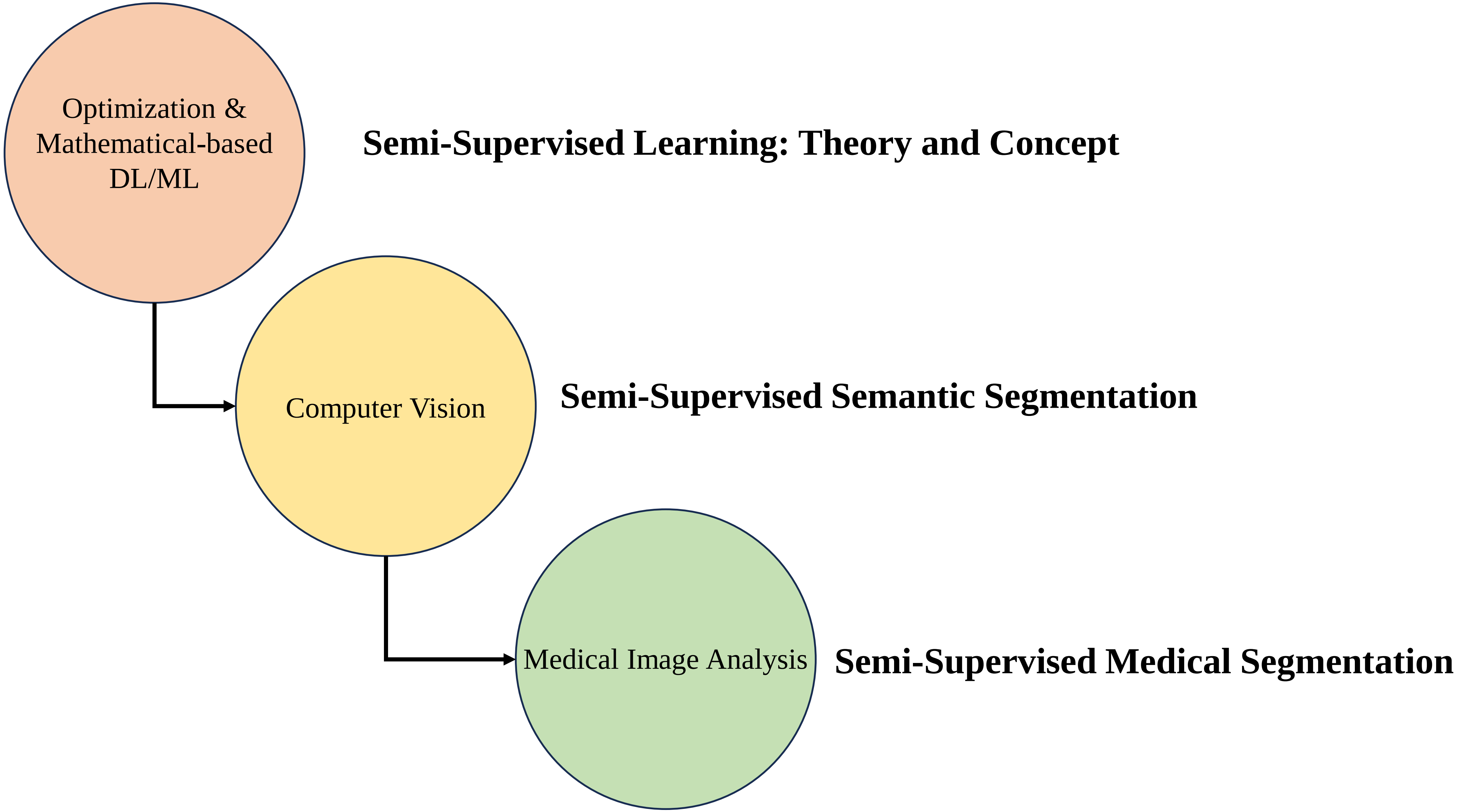
How to find your own research direction

Nguyen Thanh Huy

Knowledge Pyramid



Example: Semi-Supervised Learning

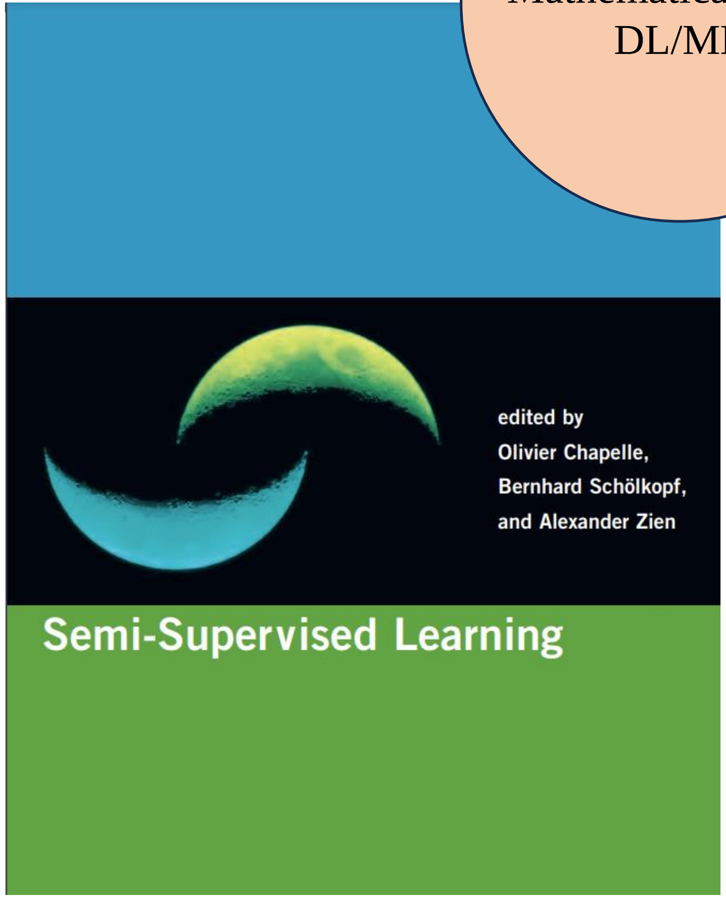


Example: Semi-Supervised Learning

Semi-Supervised Learning: Theory and Concept

Source 1 : https://www.cs.cmu.edu/~zhuxj/tmp/book.pdf
Source 2: http://www.acad.bg/ebook/ml/MITPress-%20SemiSupervised%20Learning.pdf

Optimization & Mathematical-based DL/ML

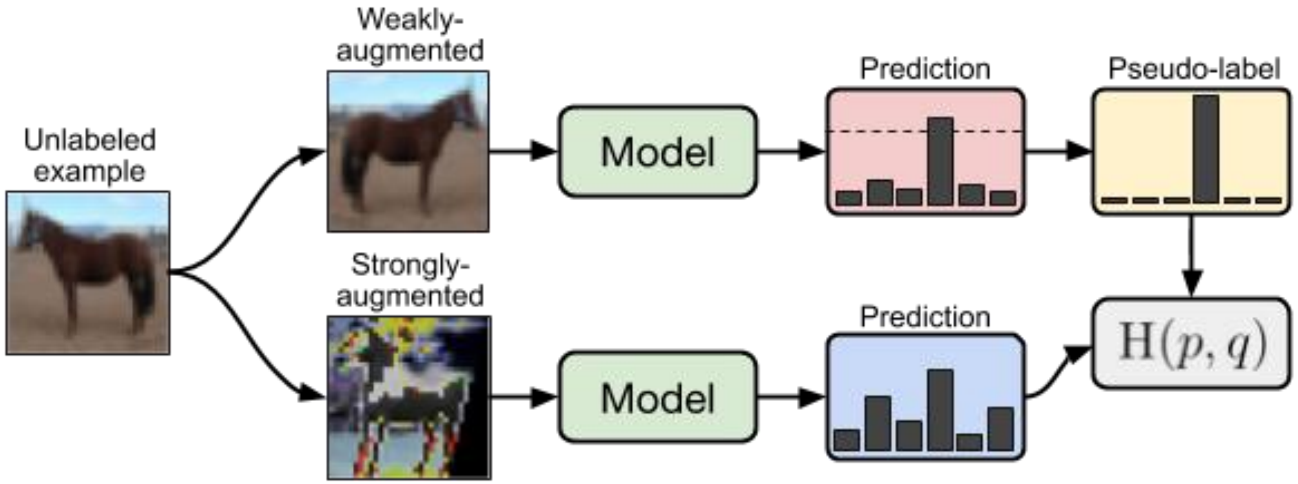


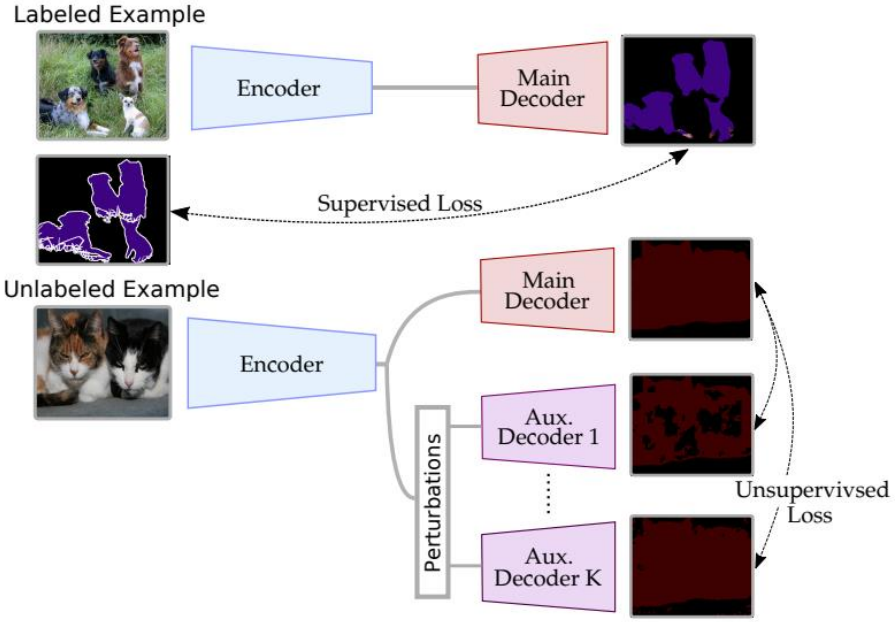
Example: Semi-Supervised Learning

Semi-Supervised Image Classification & Semantic Segmentation

Computer Vision

FixMatch: https://arxiv.org/abs/2001.07685
CCT: https://openaccess.thecvf.com/content_CVPR_2020/papers/Ouali_Semi-Supervised_Semantic_Segmentation_With_Cross-Consistency_Training_CVPR_2020_paper.pdf



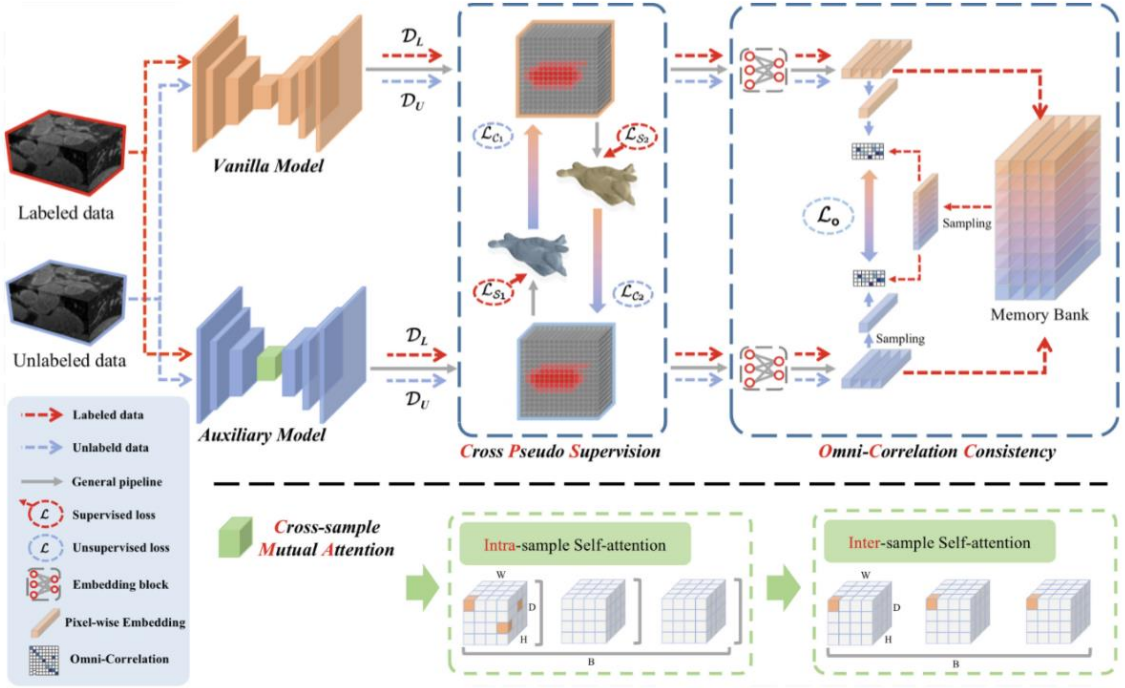


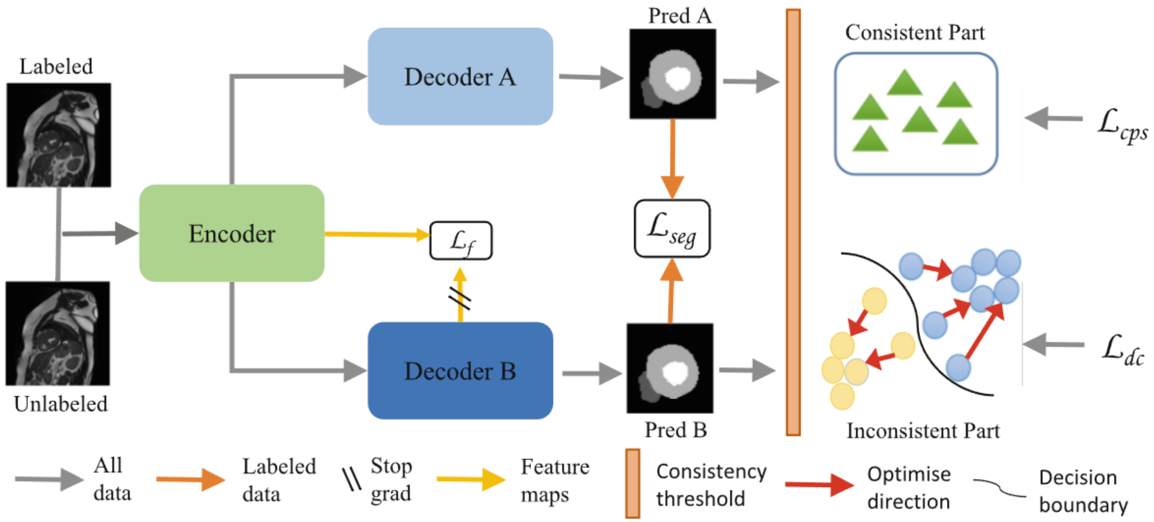
Example: Semi-Supervised Learning

Semi-Supervised Medical Segmentation

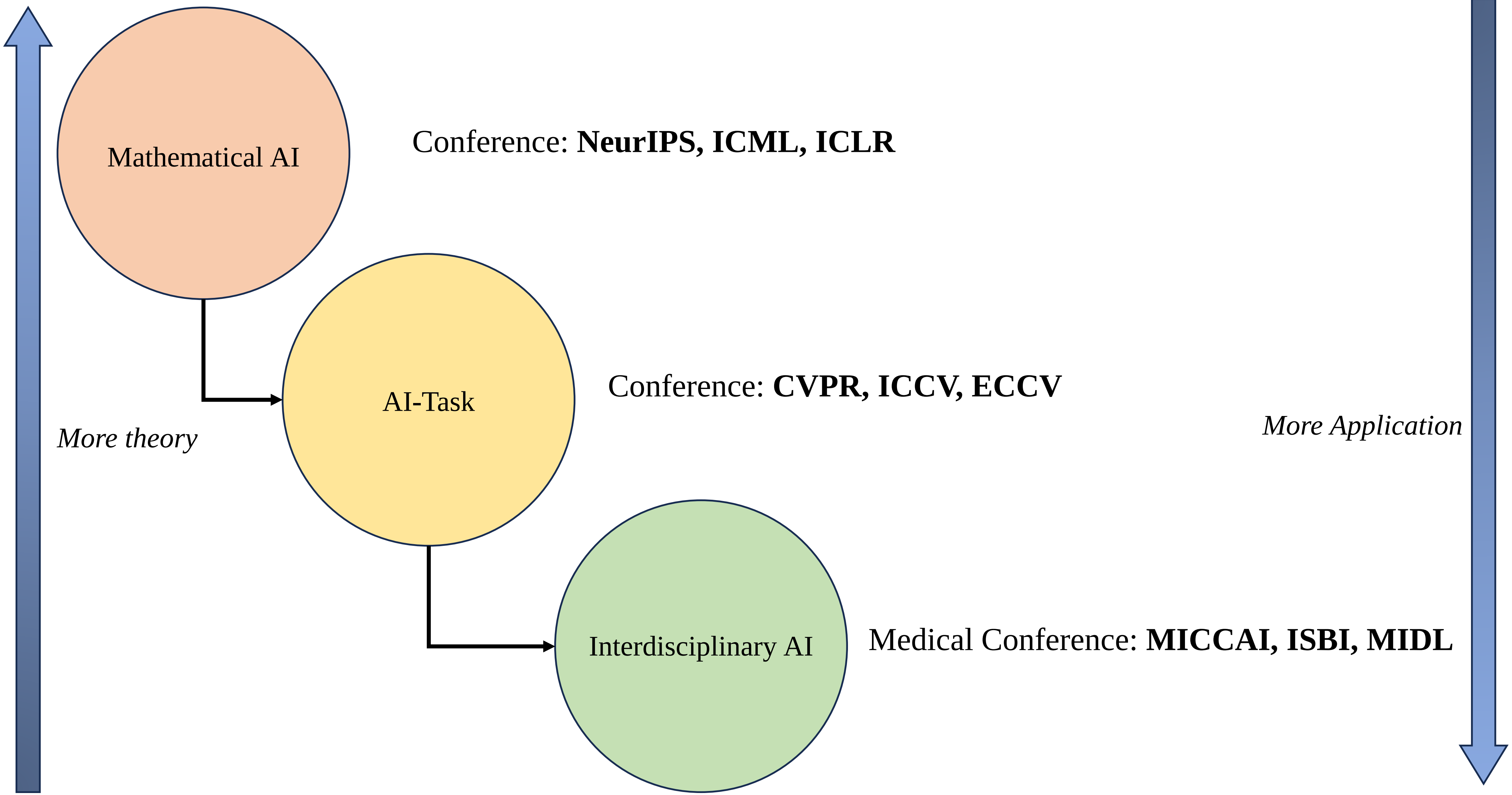
CAML: https://conferences.miccai.org/2023/papers/148-Paper0603.html
DCNet: https://conferences.miccai.org/2023/papers/171-Paper2853.html

Medical Image Analysis





AI Research Pyramid



AI Research Pyramid

Mathematical AI

Every year, there are **a lot of Mathematical AI papers** published, but only **small amount** of them is used for **AI Task**
(E.g. 10 potential related ideas for Semi-supervised Learning, but only 2 ideas are used to develop modules in Computer Vision)

AI-Task

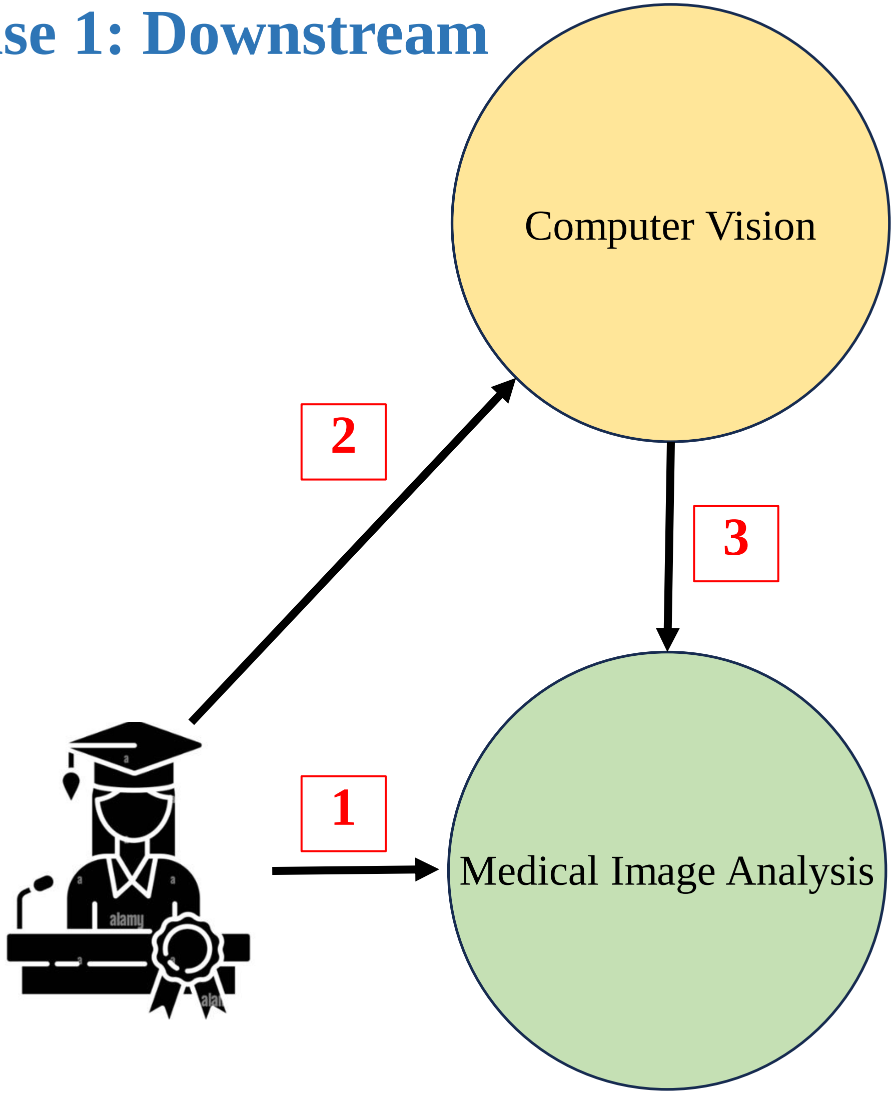
Similarly, there are **a lot of AI-task papers** published, but only **small amount** of them is used for **Interdisciplinary AI**
(E.g. 10 papers about Semi-supervised Segmentation, but only 2 ideas are used as modules in Medical Image Analysis)

Interdisciplinary AI

There's room for improvement!

How to find new ideas?

Case 1: Downstream



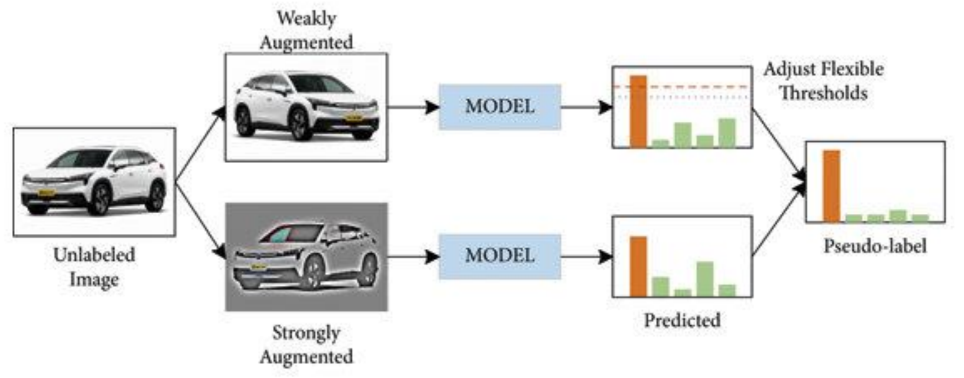
1 Do Literature Review on your Medical Image Analysis topics

2 Do Literature Review on similar Computer Vision topics

3 Make assumption and redesigned previous ideas for your topic

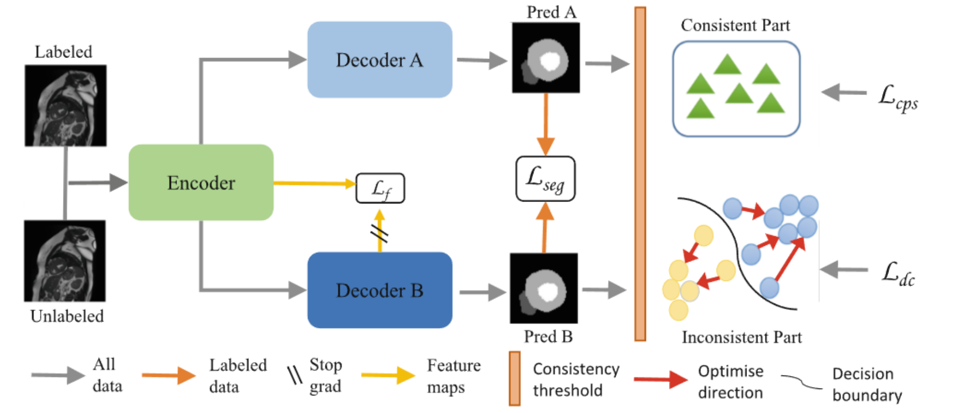
How to find new ideas?

Example 1: Downstream



2.1 Dynamic Consistency Threshold

In recent years, the pseudo labeling method based on threshold has achieved great success. The sampling strategy of this method can be defined as follows:

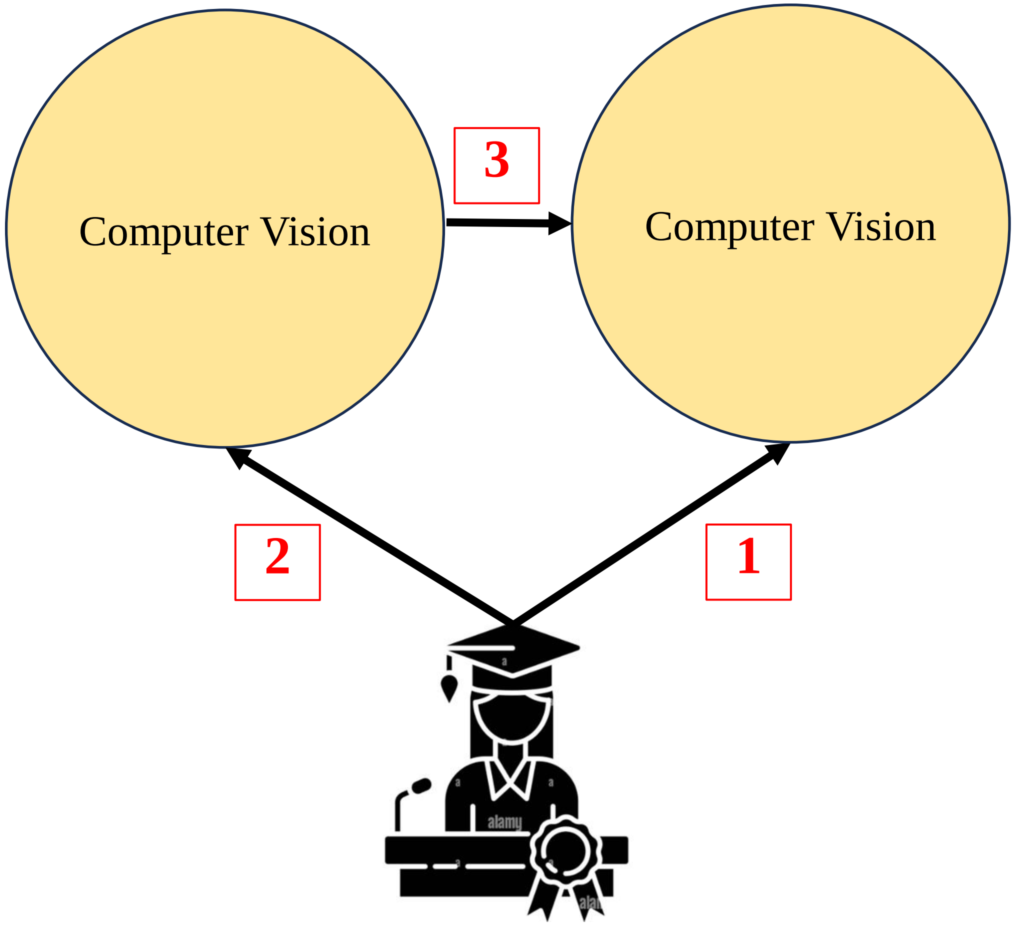


FlexMatch (2021)
Link: https://arxiv.org/abs/2110.08263

DC-Net (2023)
Link: https://conferences.miccai.org/2023/papers/171-Paper2853.html

How to find new ideas?

Case 2.1: Fill Gaps (same field)



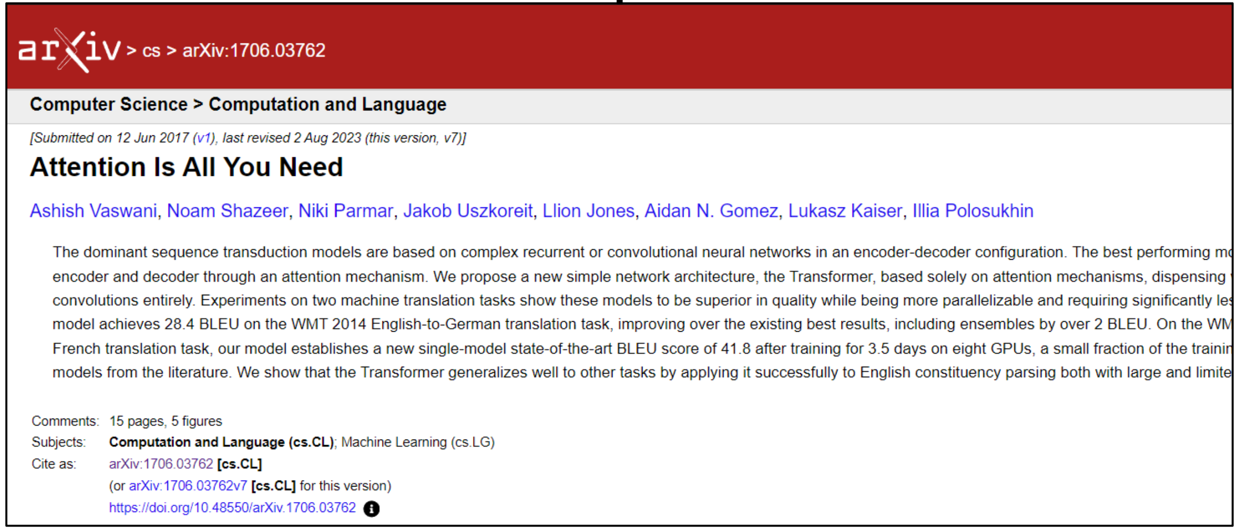
1 Do Literature Review on your Computer Vision topics

2 Do Literature Review on near/related Computer Vision topics

3 Make assumption and redesigned those ideas for your topic

How to find new ideas?

Example 2.1: Fill Gaps


arXiv > cs > arXiv:1706.03762

Computer Science > Computation and Language

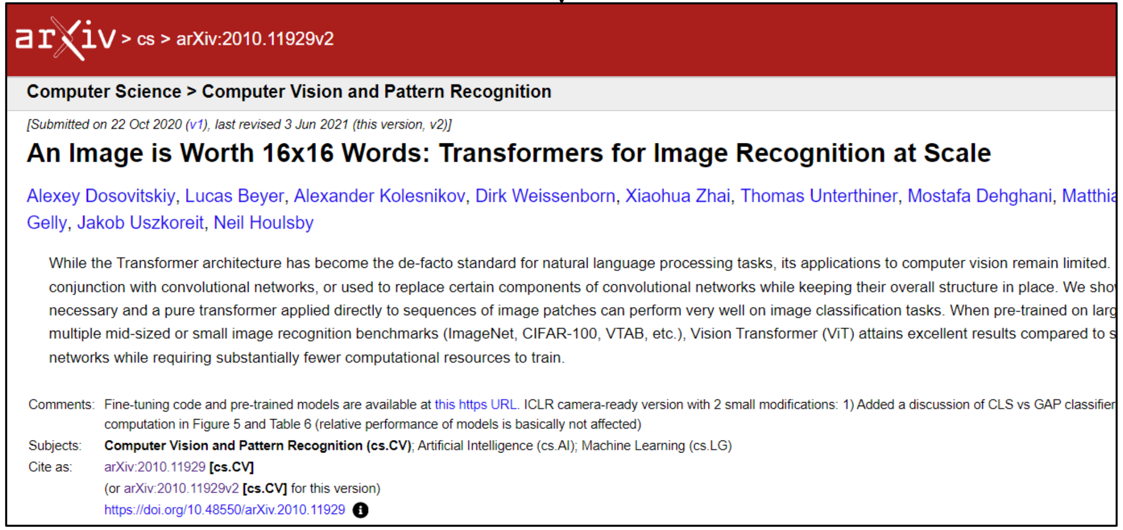
[Submitted on 12 Jun 2017 (v1), last revised 2 Aug 2023 (this version, v7)]

Attention Is All You Need

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukhin

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. The best performing mo encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly les model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles by over 2 BLEU. On the WM French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the trainin models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limite

Comments: 15 pages, 5 figures
Subjects: **Computation and Language (cs.CL)**; Machine Learning (cs.LG)
Cite as: arXiv:1706.03762 **[cs.CL]**
(or arXiv:1706.03762v7 **[cs.CL]** for this version)
https://doi.org/10.48550/arXiv.1706.03762


arXiv > cs > arXiv:2010.11929v2

Computer Science > Computer Vision and Pattern Recognition

[Submitted on 22 Oct 2020 (v1), last revised 3 Jun 2021 (this version, v2)]

An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthia Gelly, Jakob Uszkoreit, Neil Houlsby

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. conjunction with convolutional networks, or used to replace certain components of convolutional networks while keeping their overall structure in place. We sho necessary and a pure transformer applied directly to sequences of image patches can perform very well on image classification tasks. When pre-trained on larg multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision Transformer (ViT) attains excellent results compared to s networks while requiring substantially fewer computational resources to train.

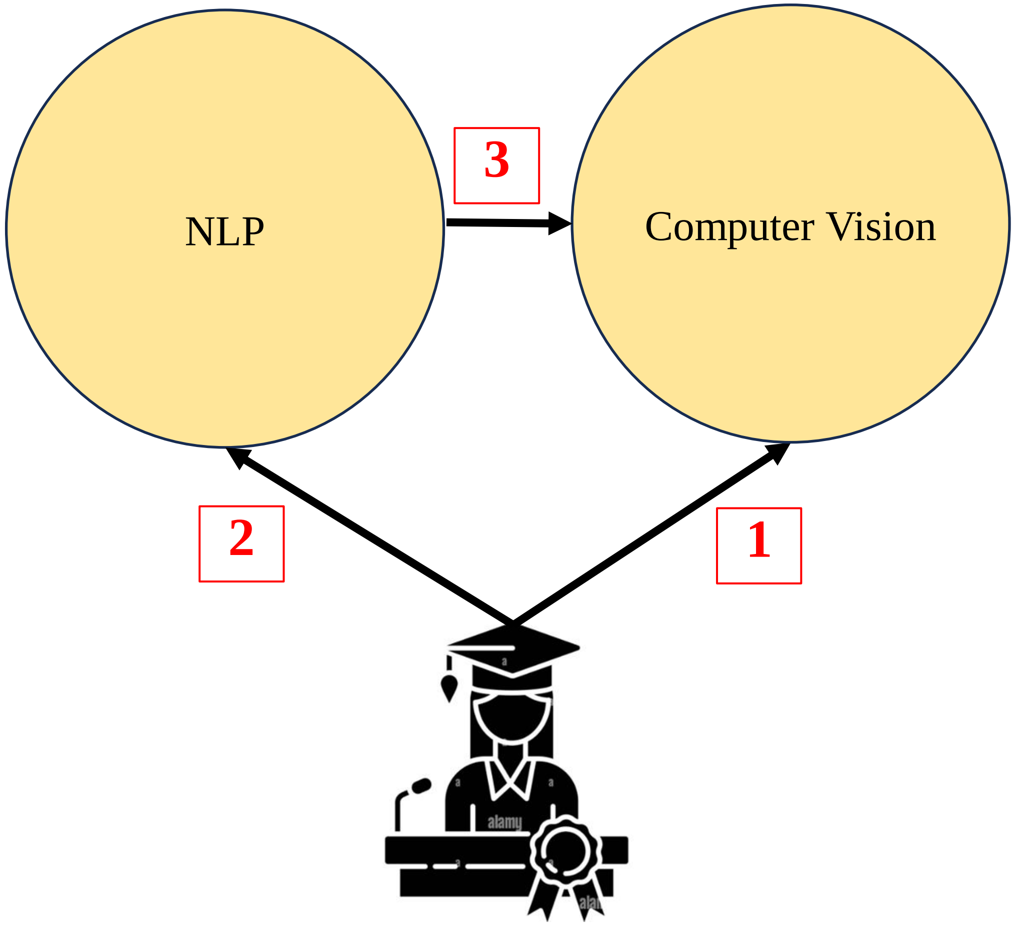
Comments: Fine-tuning code and pre-trained models are available at this https URL. ICLR camera-ready version with 2 small modifications: 1) Added a discussion of CLS vs GAP classifier computation in Figure 5 and Table 6 (relative performance of models is basically not affected)
Subjects: **Computer Vision and Pattern Recognition (cs.CV)**; Artificial Intelligence (cs.AI); Machine Learning (cs.LG)
Cite as: arXiv:2010.11929 **[cs.CV]**
(or arXiv:2010.11929v2 **[cs.CV]** for this version)
https://doi.org/10.48550/arXiv.2010.11929

Attention is all you need (Transformer - NLP) (2017)
Link: https://arxiv.org/abs/2105.05537

An Image is Worth 16x16 Words (Vision Transformer – CV) (2020)
Link: https://arxiv.org/abs/2010.11929v2

How to find new ideas?

Case 2.2: Fill Gaps (Cross-field)



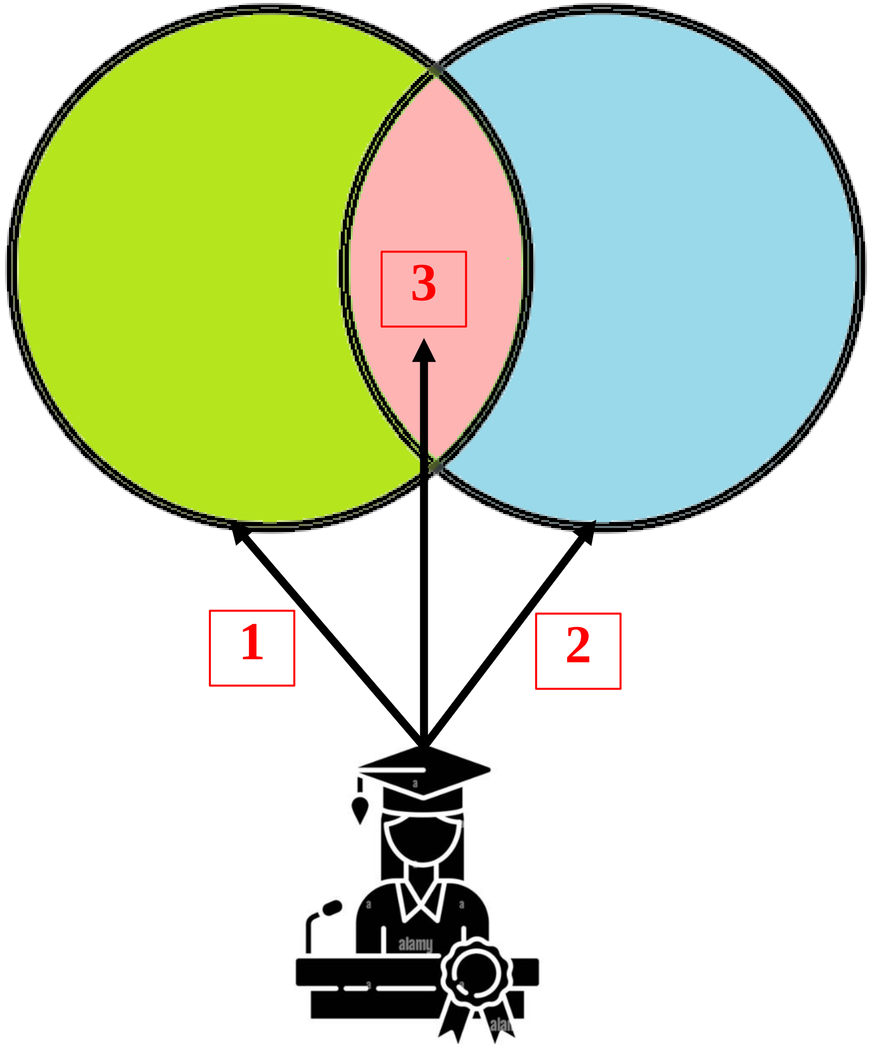
1 Do Literature Review on your Computer Vision topics

2 Do Literature Review on near/related Computer Vision topics

3 Make assumption and redesigned those ideas for your topic

How to find new ideas?

Case 3: Add Constraints



1 Do Literature Review on Topic 1

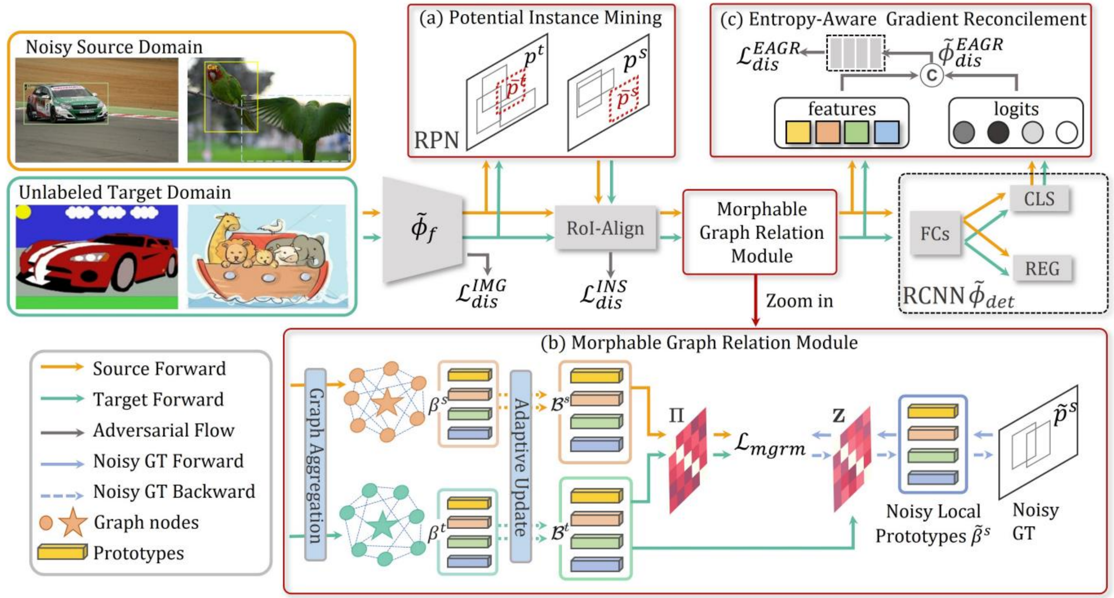
2 Do Literature Review on Topic 2

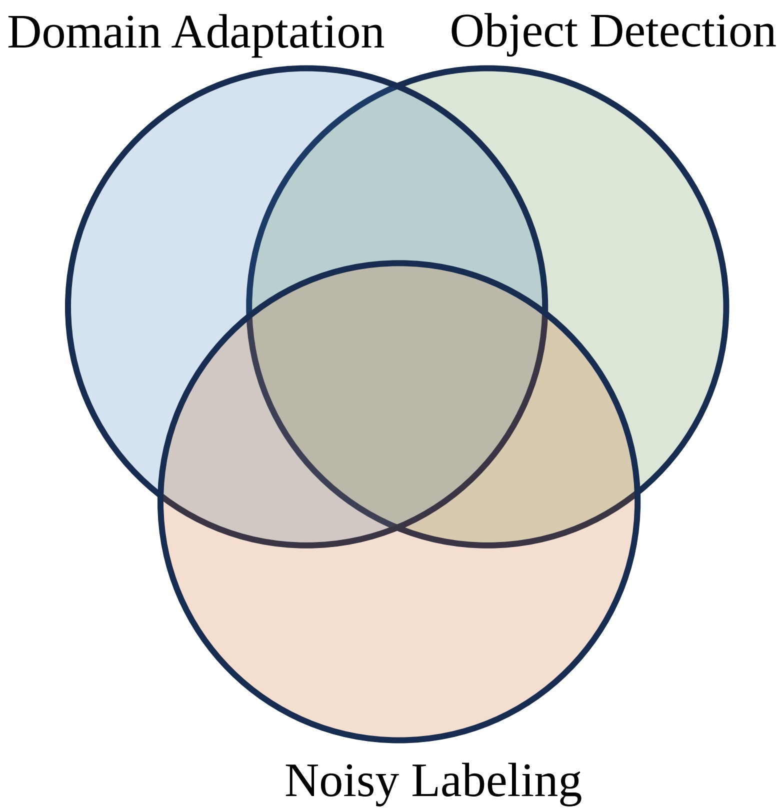
3 Redesigned Mixed ideas for your topic

How to find new ideas?

Case 3: Add Constraints

NLTE (CVPR 2022)
https://openaccess.thecvf.com/content/CVPR2022/papers/Liu_Towards_Robust_Adaptive_Object_Detection_Under_Noisy_Annotations_CVPR_2022_paper.pdf

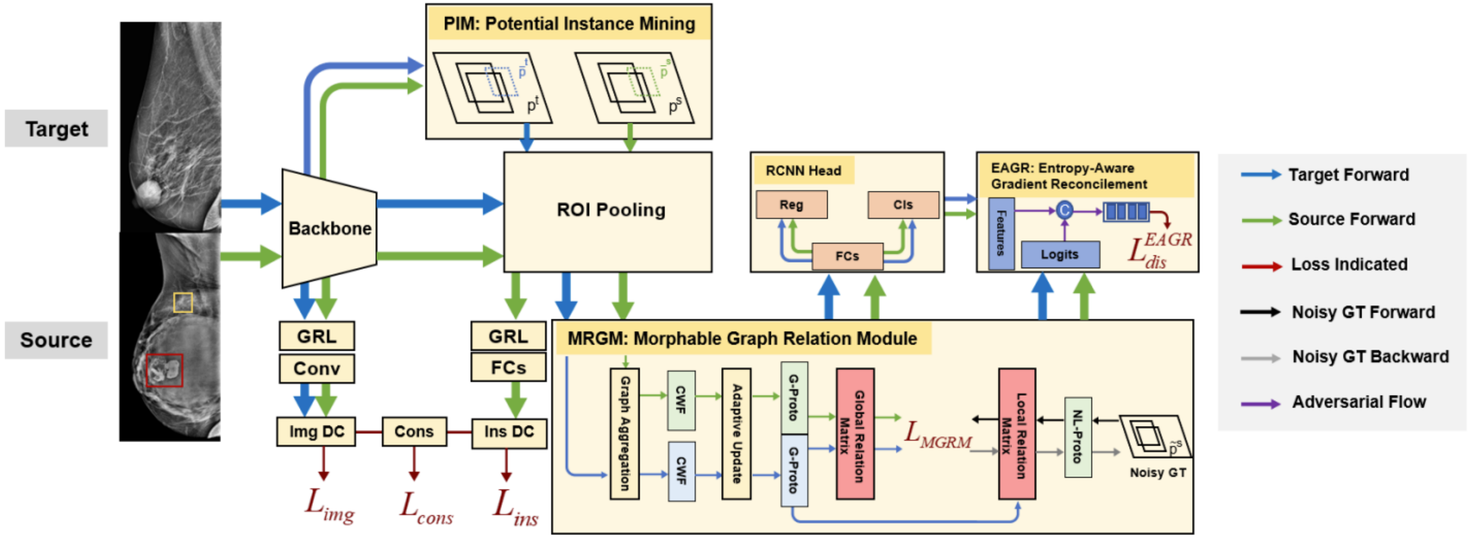


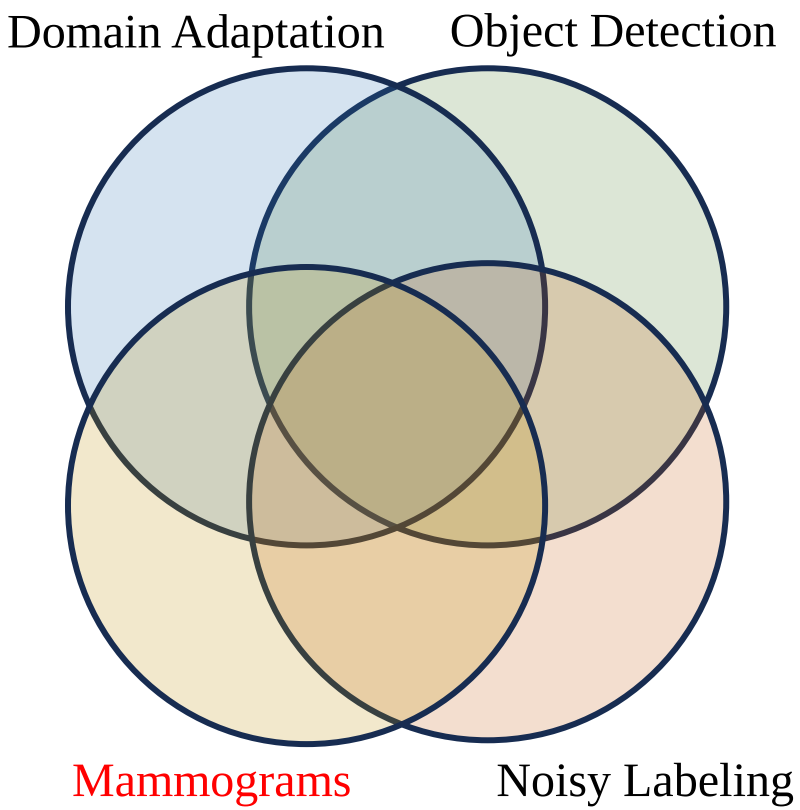


How to find new ideas?

Case 3: Add Constraints

Ours (SSP 2023)
Link: https://arxiv.org/pdf/2306.06893

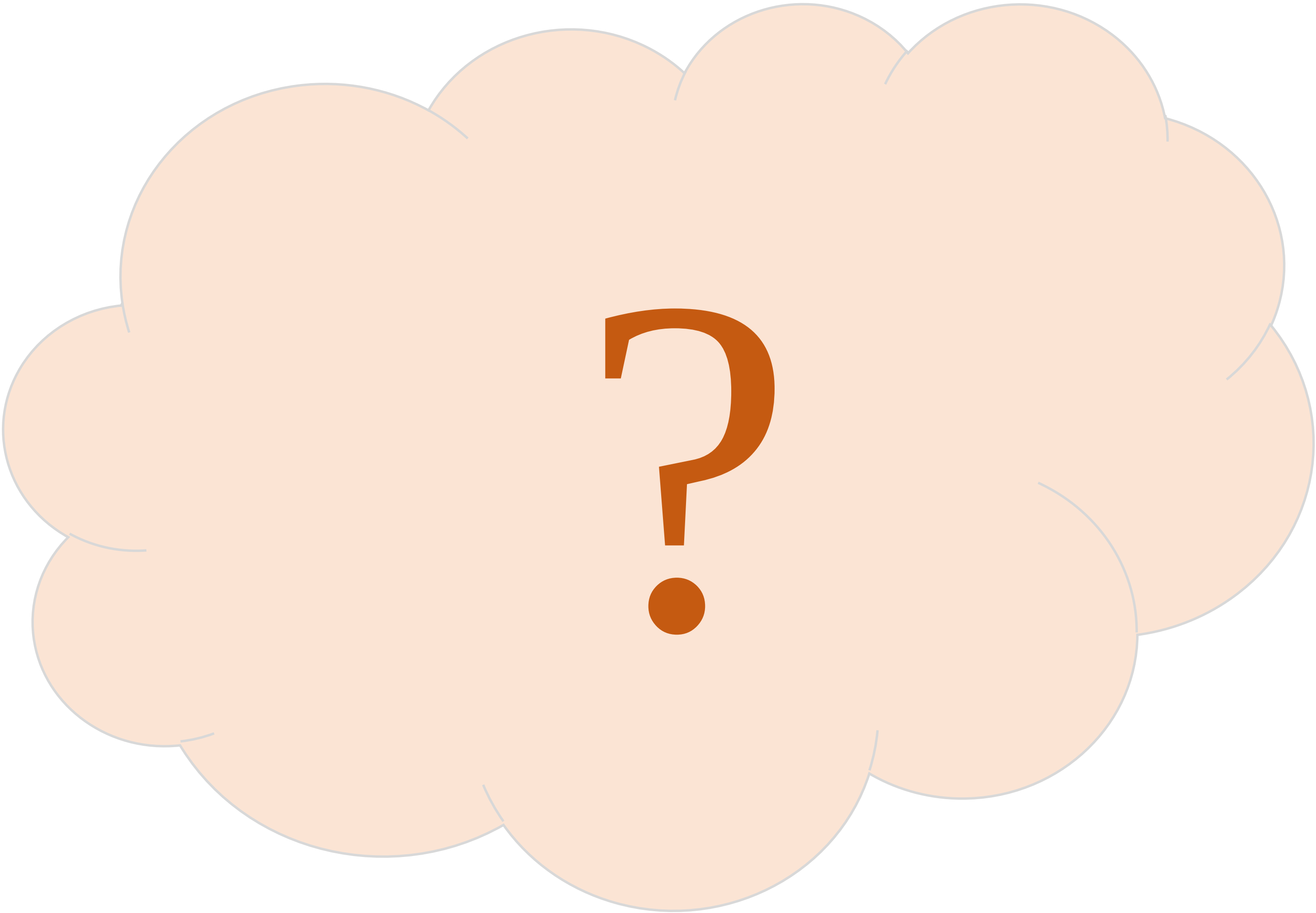




Open-End Question

Should we follow research trends?
How to identify potential topic?

Questions & Answers




Thank you!